SHARDA UNIVERSITY
Beyond Boundaries

SCHOOL OF COMPUTING SCIENCES AND ENGINEERING

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
Term 2502

COURSE CODE	COURSE TITLE	L	T	P
CSCR1503	**Application based programming in Python**	**0**	**1**	**4**

CO1: Apply fundamental programming constructs such as variables, data types, operators, conditional statements, and loops to develop basic Python programs.

CO2: Utilize Python's built-in data structures like lists, tuples, strings, sets, and dictionaries for efficient data storage, manipulation, and retrieval.

CO3: Develop modular programs using functions, including user-defined and anonymous functions, and implement exception handling for robust applications.

CO4: Implement object-oriented programming concepts such as classes, inheritance, encapsulation, and polymorphism to design reusable code structures.

CO5: Perform file handling operations and work with external modules and packages to enhance program functionality and maintainability.

CO6: Use Python libraries such as NumPy, Pandas, and Matplotlib to implement application-based solutions, including searching, sorting, and data visualization.

LIST OF EXPERIMENTS

Week	EXPERIMENT NAME	COURSE OUTCOME
1	1. Write a Python program that accepts two numbers from the user and prints their sum, difference, product, quotient, remainder and also print the datatype of each result. 2. Write a Python program to swap two numbers a) using an intermediate variable and b) without using an intermediate variable. 3. Write a program to compute distance between two points accepted from the user.	**CO1**
2	4. Write a python program to check whether a number entered by the user is positive, negative or zero. 5. An ATM contains Indian Currency notes of 100,200,500 and 2000. Write a python program which calculates and displays the minimum number of notes required to be dispensed to meet the user's requirement. 6. You are given a number A which contains only digits 0's and 1's. Your task is to make all digits the same by just flipping one digit (i.e. 0 to 1 or 1 to 0) only. If it is possible to make all the digits the same by just flipping one digit then print 'YES' else print 'NO'.	**CO1**
3	7. Write a program to accept an year as input from the user and check if the given year is a leap year or not. 8. Write a Python program to print all even numbers between 1 and 100. Also print the total count of even numbers.	**CO1**

	9. Design a Password Validation System where A user is allowed **3 attempts** to enter the correct password. Write a program that: • Asks for password input • Stops when the correct password is entered or attempts are exhausted. correct_password = "python123"	
4	10. Write a Python program to concatenate two lists and then perform element-wise multiplication of the resulting list by 3. 11. You are provided with a list of grocery items purchased by a customer. As shown below. Write python code to print the name of the item and its quantity (occurrences in the list). L=[“pen”,“ball”,“eraser”,“pen”,“band”,“pen”,“Pencil”,“ball”] Which item is the most occurring item from the list?" 12. Given a nested list, extract and print the last element of each inner list using list indexing.	**CO2**
5	13. Given a list of tuples representing books with (title, author, publication year), create a Python function that sorts the list based on the publication year in ascending order. Return the sorted list. 14. Create a Python program that initializes a tuple of tuples, where each inner tuple represents a student with (name, age, grade). Sort the students based on their grades in descending order and return the sorted list. 15. You are provided with following data consisting of grocery items and their cost. price = [('item1', '17.20'), ('item2', '15.10'), ('item3', '24.5')] sort the above data on the basis of price and display it in order. Which item is the second costliest in the list.	**CO2**
6	16. Write a python program that takes a string as input and returns a new string where each word is reversed. For example, if the input is "Hello World," the output should be "olleH dlroW." 17. Write a Python function that takes a list of strings as input and returns a new list containing only the strings with more than five characters. Additionally, reverse the characters in each string. 18. Write a Python program to check the strength of a password entered by the user. Implement conditions to check for the following: Length of at least 8 characters. Presence of both uppercase and lowercase letters. Presence of at least one digit. Presence of at least one special character (!@#$%^&*).	**CO2**
7	19. Design a Python script that represents a simple student database using a dictionary. Allow the user to add new students, update their grades, and retrieve the average grade for all students in the database. 20. You are provided with the following data which contains details of few brands of mobile phones. l=[{'make':'Samsung','model':'j2','year':2016}, {'make':'Xiomi','model':'note3','year':2018}, {'make':'Apple','model':'XE','year':2019}] Write a program to sort this data on the basis of attribute ‘make’ and print it." 21. You are given with a list of dictionaries having keys (name, marks). Write a program to filter the name of students on the basis of marks value given by user. Example display names of all the students having marks >95.	**CO2**
8	22. Write a program to calculate simple interest. Suppose the customer is a senior citizen. He is being offered 12% rate of interest: for all other customers ROI is 10%. 3. Define an anonymous function (lambda function) that squares a given	**CO3**

	number. Use the lambda function to square the numbers 1 to 5 and print the results. 23. Write a recursive function to generate the first n terms of the Fibonacci sequence. 24. Using map() and lambda expression write a program to convert the following list of temperatures to Fahrenheit. Celsius=[34.56,23.67,45.67,67,34]	
9	25. Design a Python function that takes two parameters (numerator and 26. denominator) and calculates the quotient. Handle potential division by zero exceptions using a custom exception. Test the function with different inputs. 27. Define a function named greet_user that takes two parameters, name and greeting. If greeting is not provided, the function should default to "Hello." The function should then print the greeting followed by the name. 28. Write a number game program ask the user to enter your number. If number is greater then number to be guessed , raise a ValueTooLarge exception. If the value is smaller than the value to be guessed , raise a ValueTooSmall exception and prompt the user to enter again. Quit the program when the user enters the correct number.	**CO3**

10	29. For the following class
	class Point:
	def __init__(self,x,y):
	self.x=x
	self.y=y
	a. overload (-) operator to calculate distance between points P1 and P2.
	b. overload (>=) operator to check if one point is at a more distance from origin than the other.
	
	30. Write a program using the concept of classes, objects, and exception handling to accept the details of a Class XII student applying for a professional course.
	The program should accept: Student Name, Roll Number and Marks in Physics, Chemistry, and Mathematics
	
	If the user enters marks less than 0 or greater than 100 for any subject, a user-defined exception should be raised and handled appropriately. Display the student details and calculate the average percentage if inputs are valid.
	
	31. Write a Python program to model a Bank Account Management System using following features:
	a) Create an abstract class Account with abstract method calculate_interest().
	b) Implement private data members for account details.
	c) Derive SavingsAccount and CurrentAccount classes.
	d) Override calculate_interest() in derived classes.
	e) Create objects and demonstrate runtime polymorphism.
		CO4

11	32. Solve the following system of linear equations using the linalg module of NumPy. Also check correctness of the solution. $2x + 3y + z = 13$ $x - y + 2z = 7$ $3x + 4y + z = 22$	**CO5**

	33. Create a Pandas Series and display statistical details. (mean, max, min)	
12	34. You are given a CSV file named `student.csv', whose first few records are as below. Write the python code / command/ expression for following questions Name Age Sex Marks Grade Nihar 23 M 23 C Ranjan 34 M 40 B Roy 23 M 47 A Sunita 18 F 49 A a. Load this file into a python DataFrame b. Find the number of rows and columns in it. c. See last 5 rows from the bottom d. Find the details of student with lowest marks e. Find sum total marks of all female students f. List names of all the male students g. Find mean age of the class h. Line plot marks of the class i. Find the index of record of oldest student in the class J. Who are better students 'Males' or 'Females'. Justify on the basis of above data. 35. Plot a comparison plot of following two expressions y = x2 + 2x z = sin(y) 36. Assume that you are given marks of 60 student of Python course between 0-20. Plot a histogram from this. 37. Write a Python program to Analyze monthly sales data using Pandas. Determine the total sales and highest selling month. Also plot a pie chart of monthly expenses.	**CO5**
13	38. Write a Python program to Implement Linear Search algorithm. 39. Write a Python program to Implement Binary Search algorithm. 40. Write a Python program to Implement Bubble Sort algorithm.	**CO5**

VALUE ADDED EXPERIMENTS

S.NO.	EXPERIMENT NAME	CO
1	Students will take up a mini project in a group of not more than 3 students, and present it at the end. The project title and statement needs to be approved by the teacher	**CO6**

Course Coordinator **Lab Coordinator** **Head of the Department**